INCH-POUND

MIL-C-24643/55D
22 November 1994
SUPERSEDING
MIL-C-24643/55C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS4NW8
(INCLUDING VARIATION LS4NWA8)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS4NW8 (unarmored) and LS4NW8 (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, AWG size 8, class B stranding.
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Cross-linked polyethylene insulation or ethylene propylene rubber insulation. Minimum average thickness 0.045 inch. Standard identification code shall be applied by method 1 or method 3. When identification method 1 is used, the insulation color shall be black.
Fourth - Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LS4NW8

Sixth - Cross-linked polyolefin jacket. Minimum average thickness 0.050 inch. Maximum overall diameter 0.74 inch.

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REQUIREMENTS (Continued)

Construction variant for type LS4NWA8

Sixth - Cross-linked polyolefin jacket. Minimum average thickness 0.050 inch Maximum overall diameter 0.79 inch.
Seventh- Braided metal armor.

Notes: 1. Variant LS4NW8 military part number - M24643/55-01UN
2. Variant LS4NW8A military part number - M224643/55-02AN

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	0.679
Voltage withstand - volts, root mean square, minimum	4500
Insulation resistance - Conductor to conductor Megohms/1000 feet minimum	4200
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C)	50
Cold bending, cable - 6 inch mandrel diameter maximum	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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EXAMINATION AND TESTS (Continued):	Requirements
Group C:	
Physicals - (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) cycles, minimum (method 1 only)	25
* Permanence of printing (jacket) - cycles, minimum	125
Armor - Type LS4NWA8 only, conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ±2°C)	No damage
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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QUALIFICATION INSPECTION (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1500 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
Army - AV, CR, ER, ME, AR, AL
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